Sophia Guarany Cabral Casino

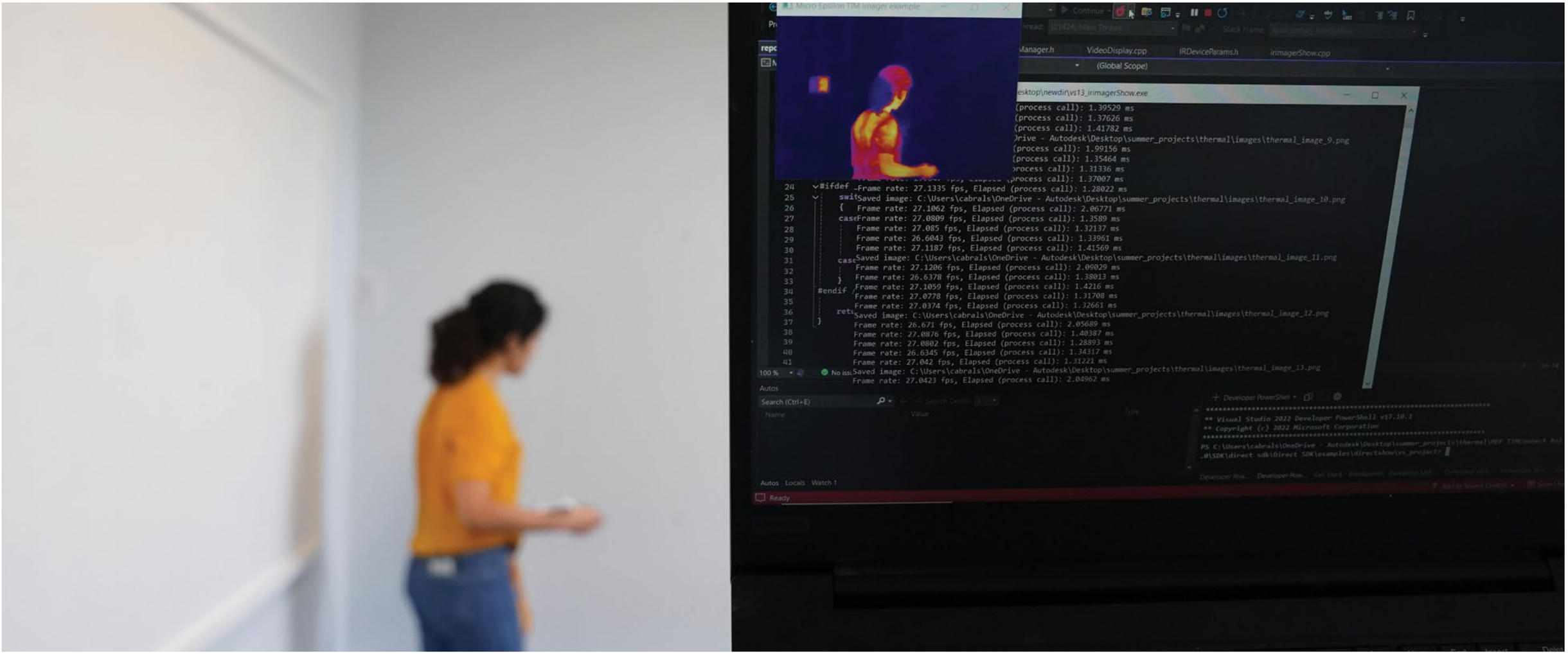
This portfolio is a compilation of my academic and professional work, observations, and explorations.

Recent + On-going work:

A Contactless Multi-Modal Sensing Approach for Material Assessment and Recovery in Building Deconstruction:

This study introduces an innovative system combining multi-modal sensing technologies and machine learning to assess building materials for reuse potential, addressing growing environmental concerns. The system integrates thermal imaging, RGB cameras, and depth sensors to analyze material conditions and hidden geometries within existing buildings, enhancing understanding of materials and assemblies.

Link: https://www.mdpi.com/2071-1050/17/2/585#



Type: Research paper
Research Collaborators: Mikita Klimenka, Fopefoluwa Bademosi, Damon Lau, Stefanie Pender, Lorenzo Villaggi, James Stoddart, James Donnelly, Peter Storey, and DavidBenjamin
Year: 2024

Awards and Recognition:
Accepted article under MDPI "Sustainability" as part of the Special Issue "A Circular Economy for a Cleaner Built Environment"
Selected to present at the 23rd Annual New York State Green Building Conference.

Localized Biocomposite 3D-Printing with Natural Fibers:

A research exploring a framework for the manufacturing of e 3D printable biocomposites made from globally available agricultural fibers, leading to five regionally specific biocomposites. Each biocomposite includes a natural fiber blended with two universally available additives - a biodegradable binder and cellulose.

Link: https://papers.cumincad.org/data/works/att/acadia24_v2_95.pdf

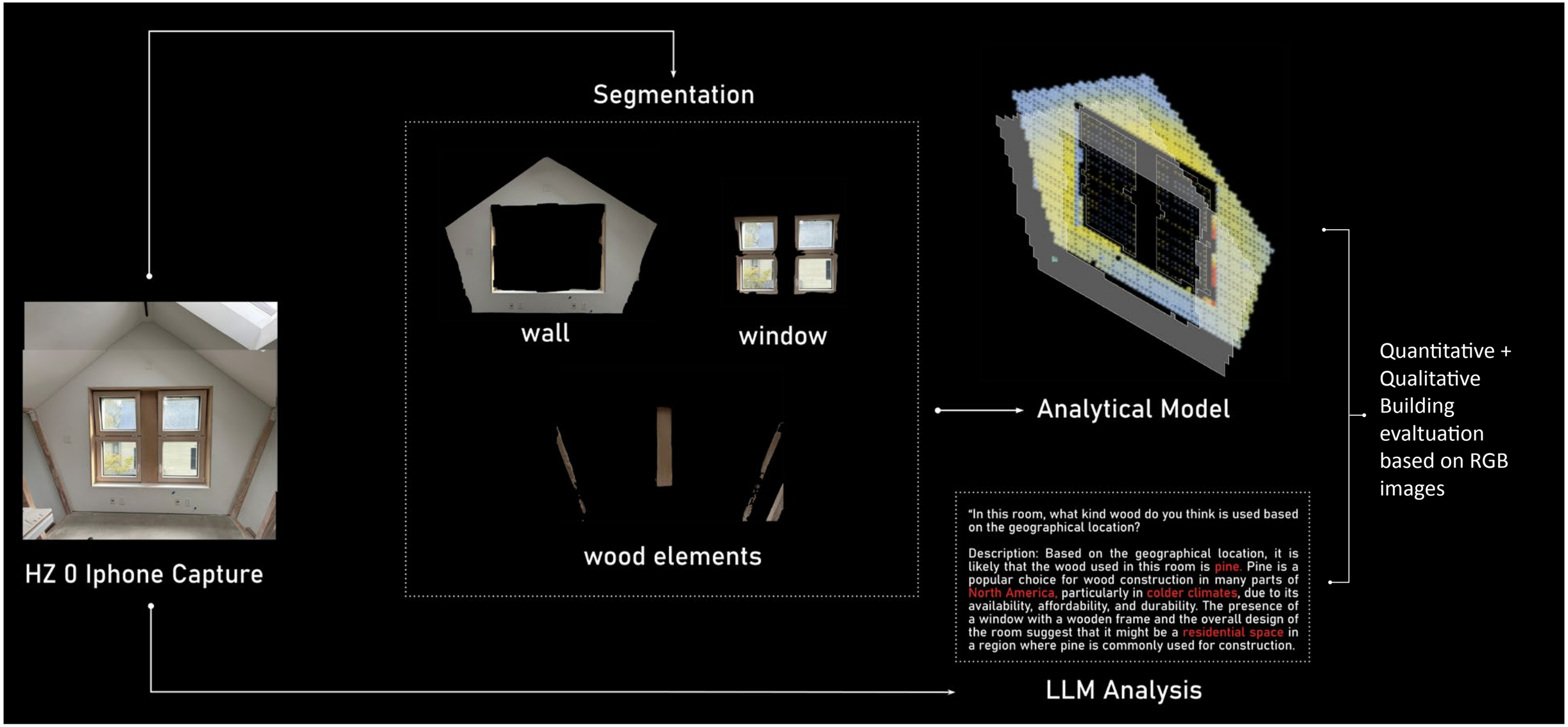
Type: Research paper
Research Collaborators: Daniel Tish , Amelia Wen Jiun Gan, Martin Bechthold (Harvard Affiliation)
Year: 2024 - Present

Awards and Recognition: ACADIA Conference Runner-Up for the Vanguard Award.

Project VIS: Image-Based Insights and Predictions for the Built Environment:

Ongoing research at Grinham Research Group, funded by Autodesk, explores explores what interior conditions can be inferred from building images. Two methods are evaluated: an image regression algorithm that predicts environmental conditions and a retrieval-augmented generation approach that uses a fine-tuned language model with contextual building information.

Link: https://research.gsd.harvard.edu/grg/publications/



Type: Research paper
Research Collaborators: Jonathan Grinham (Harvard affiliation)
Year: 2024 - Present

Status:
Currenlty under testing and findings stage.

Harvard Teaching and workshops:

1-Initialization Code Example:

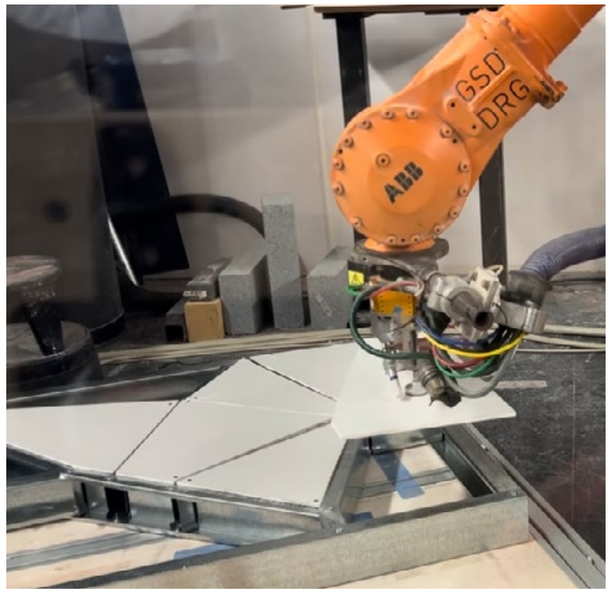
```
                    ; Start Code for Dremel 3D45 Printer
M140 S45            ; Set the heated bed temperature to 45°C (start heating)
M105                ; Report current temperature of both the extruder and the heated
                    bed
M190 S45            ; Wait for the bed temperature to reach 45°C
M104 S220           ; Set extruder temperature to 220°C (start heating)
M105                ; Report current temperature of both the extruder and the heated
                    bed
M109 S220           ; Wait for the extruder to reach 220°C
G28                 ; Home all axes (X, Y, Z) to the printer's origin
G1 Z50.00 F400      ; Move the Z-axis to a height of 50mm at a speed of 400 mm/min
G92 E0              ; Reset the extruder position to zero
G1 F200 E3          ; Extrude 3mm of filament at a speed of 200 mm/min
G92 E0              ; Reset the extruder position to zero again
M132 X Y Z A        ; Store the current motor positions for X, Y, Z, and A (extruder
                    axis)
M907 X100 Y100 Z50 A100 ; Set motor current values: X, Y = 100%, Z = 50%, A (extruder)
                    = 100%
G92 E0              ; Reset the extruder position to zero
G92 E0              ; Reset the extruder position to zero (redundant command, possibly
                    for extra safety)
G1 F1800 E-1.9      ; Retract 1.9mm of filament at a speed of 1800 mm/min (preparing
                    for print start)
```

2-Printing Code Example "Core Code":

```
G1 F300 X6.549 Y4.112 Z0 E0.01283
G1 F300 X6.294 Y4.135 Z0 E0.02566
G1 F300 X6.038 Y4.159 Z0 E0.03848
G1 F300 X5.783 Y4.182 Z0 E0.05131
G1 F300 X5.527 Y4.206 Z0 E0.06414
G1 F300 X5.272 Y4.229 Z0 E0.07697
G1 F300 X5.016 Y4.252 Z0 E0.0898
G1 F300 X4.764 Y4.28 Z0 E0.10251
G1 F300 X4.641 Y4.505 Z0 E0.11534
G1 F300 X4.518 Y4.73 Z0 E0.12816
G1 F300 X4.395 Y4.956 Z0 E0.14099
G1 F300 X4.272 Y5.181 Z0 E0.15382
G1 F300 X4.149 Y5.406 Z0 E0.16665
G1 F300 X4.024 Y5.628 Z0 E0.17939
G1 F300 X3.767 Y5.628 Z0 E0.19222
```

Computational Design TA

+ Custom G-code Workshops



Large Scale Prototyping TA

+ Robotic arm assembly Workshops

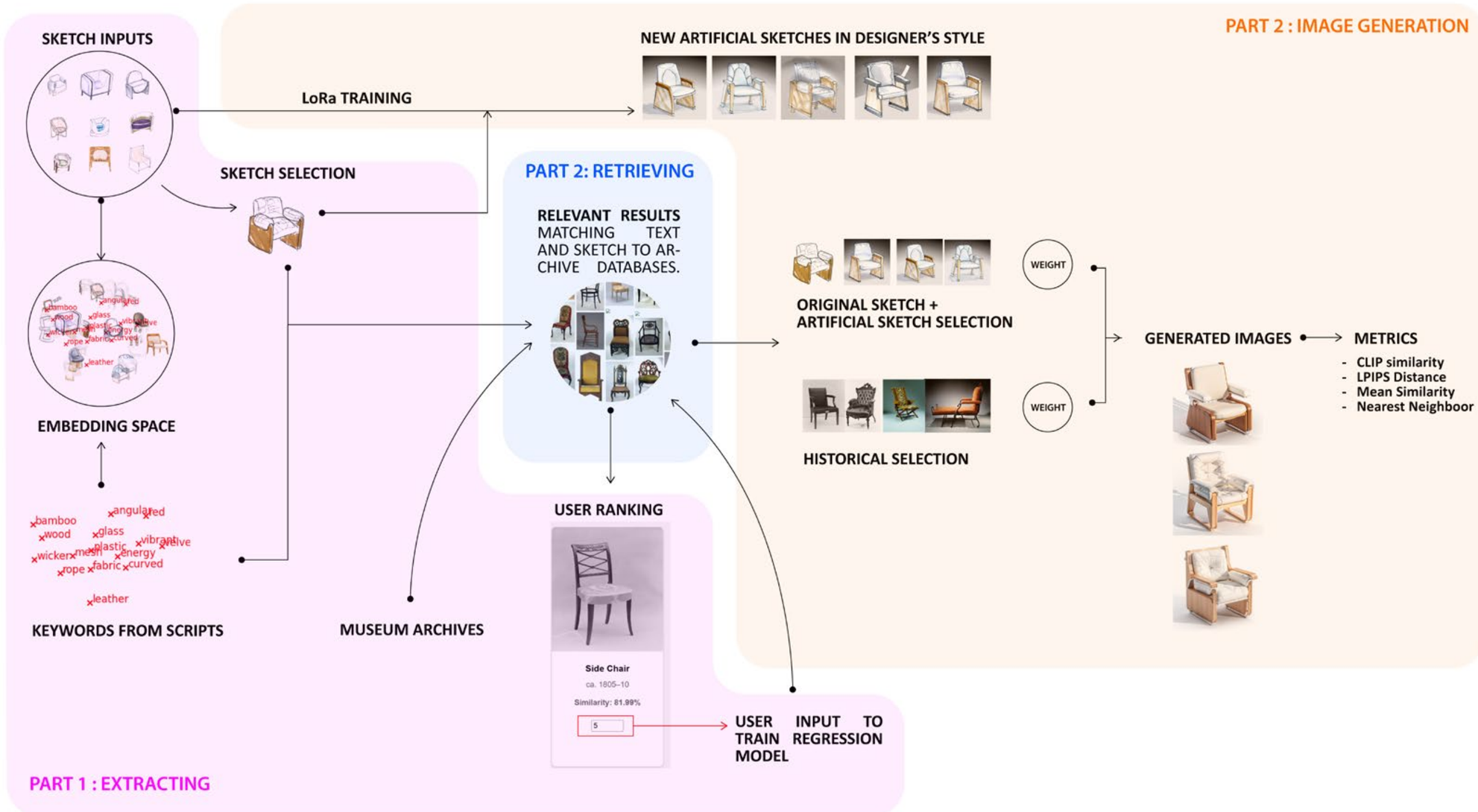
```
# Total heat loss
total_heat_loss = heat_loss_wall + heat_loss_window

# Calculate percentages only
percent_wall_loss = (heat_loss_wall / total_heat_loss) * 100 if total_heat_loss >
percent_window_loss = (heat_loss_window / total_heat_loss) * 100 if total_heat_lo

# Print only percentage loss
print(f"Percentage of Heat Loss from Walls: {percent_wall_loss:.2f}%")
print(f"Percentage of Heat Loss from Windows: {percent_window_loss:.2f}%")
```

Circular practices TA

+ Machine Learning for Architects workshops

Sketching Memories: A Machine Learning Framework for Enhancing Design Ideation with Sketches, Words, and Visual Archives:

This research paper and tool from the MIT course AI and ML for Engineering Design explores a context-aware design process using computer vision, NLP, and generative AI. It leverages CLIP embeddings and museum databases to extract emotional cues, retrieve related imagery, and generate new design iterations. Currently in development for publication.



Type: Research paper and digital tool
Research Collaborators: Faez Ahmed (MIT affiliation) and Sarah Higgins (Anthropologie)
Year: 2024 - Present

Status:
Currenlty under testing, findings, and finetuning stage.

MicroBot:

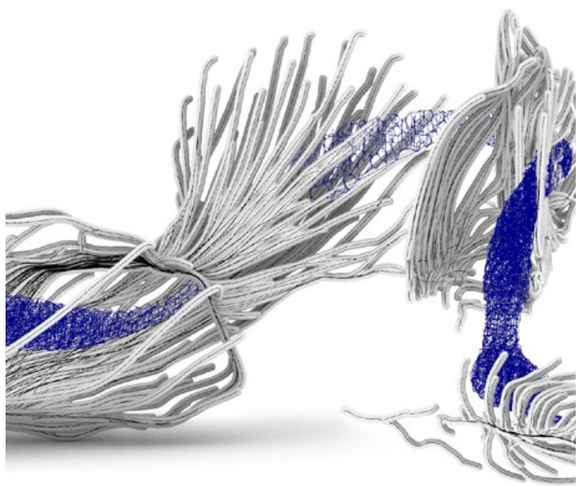
Link to short clip

Type: Research and exhibition
Research Collaborators: Allen Sayegh (Harvard affiliation)
Year: 2025

Status:
MDes Open Projects Completion

Past Work + Background:

Previous M.Arch Experience:



Parametric workflows +

Organic Geometries

Link

Exploring robotic arm extrusions workflows through early-stage training and trials.

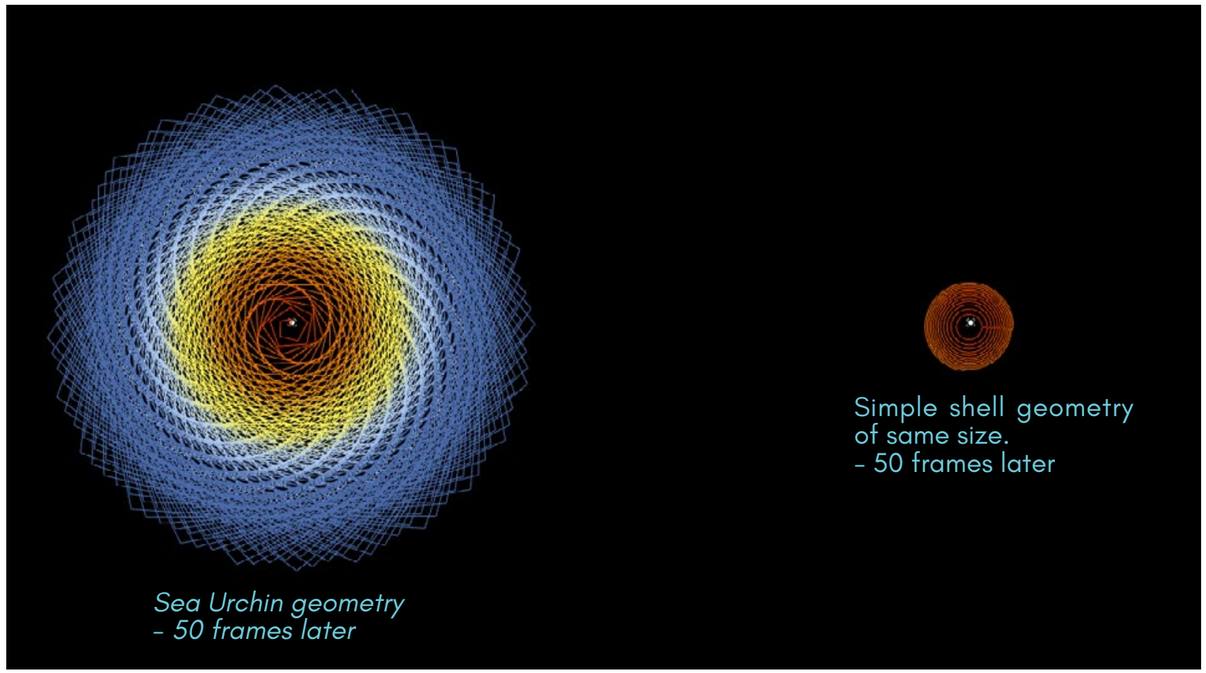
Development and delivery of robotic arm training modules through the Robotics Academy, funded by a National Science Foundation (NSF) grant.

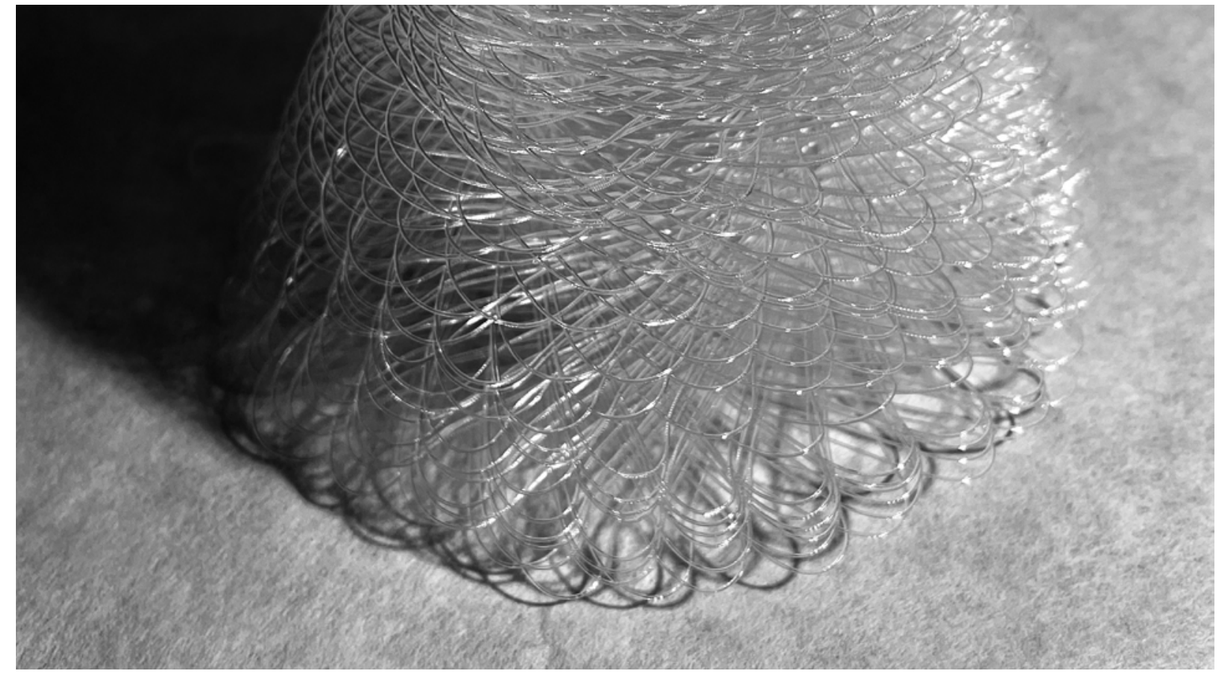
Link

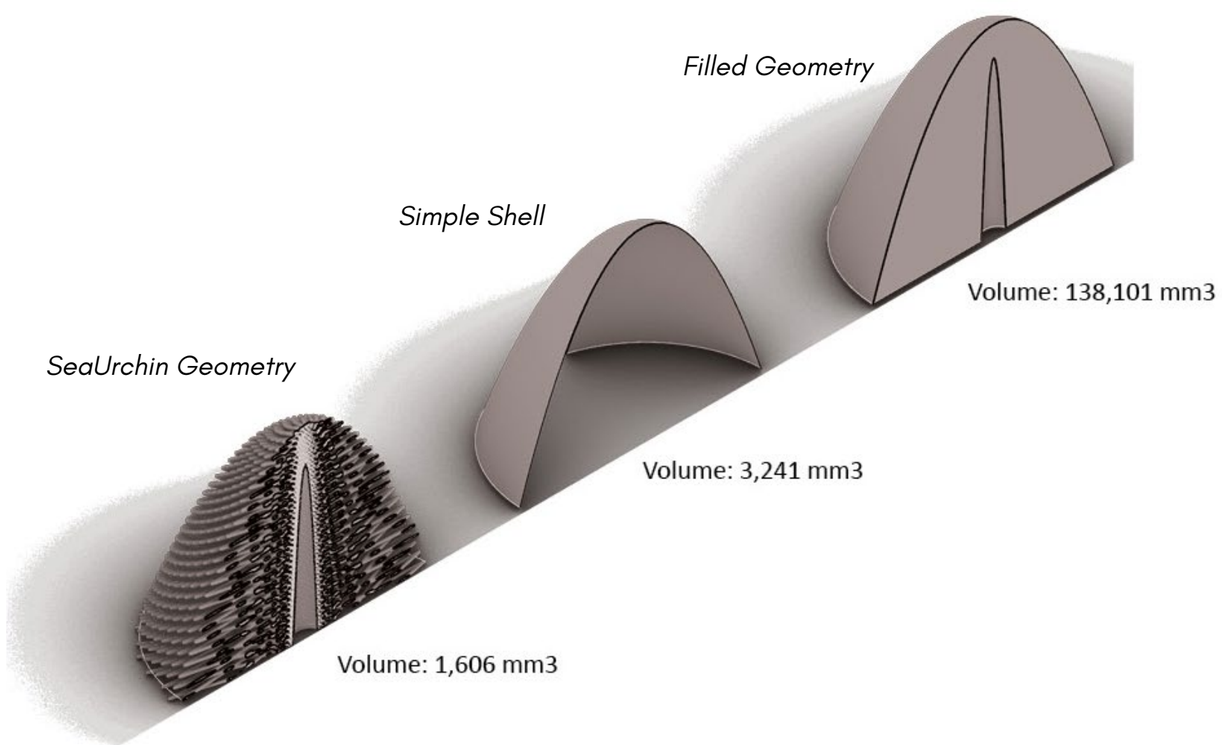
For fun:

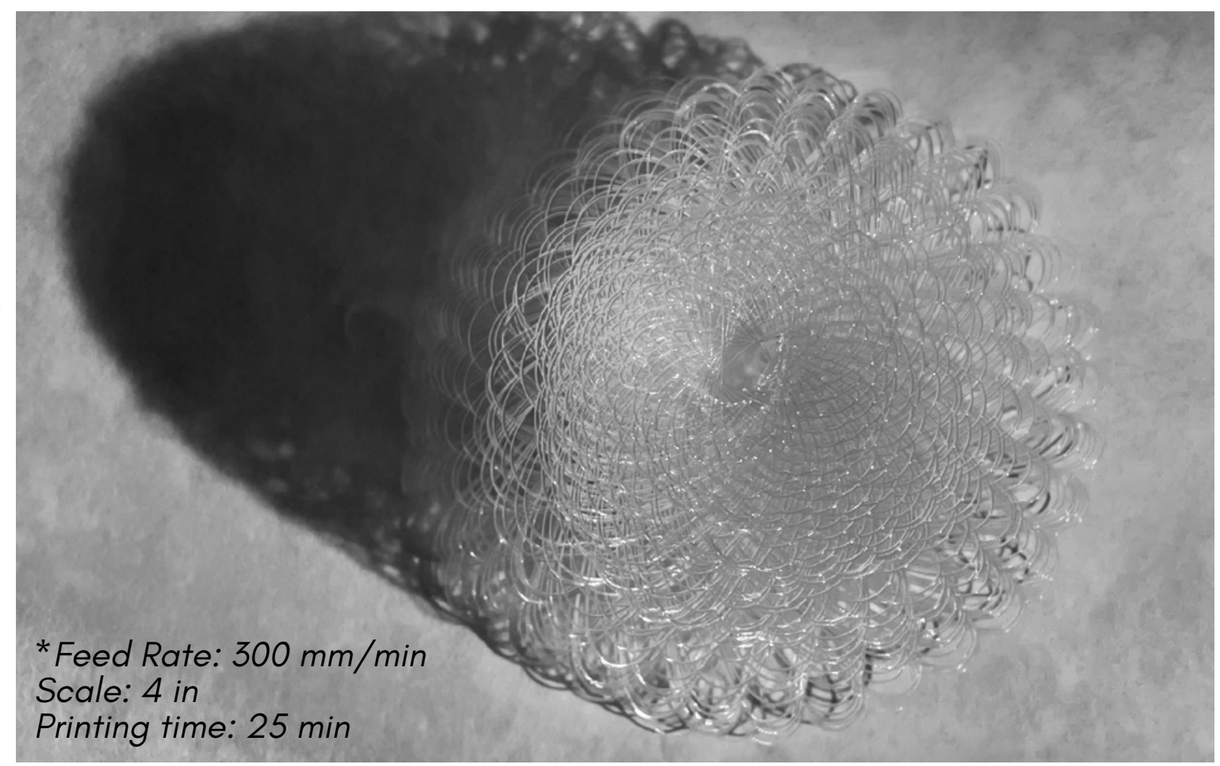
Concept + Introduction:

SeaUrchin explores the numerous possibilities of phyllotaxic algorithms to create intricately detailed forms quickly and with minimal volume. The plug-in ensures proper layer heights for 3D printing purposes, and designers can achieve various shapes depending on how they set their own feed rates and geometry scales.



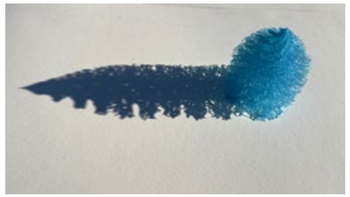


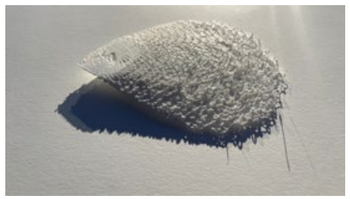


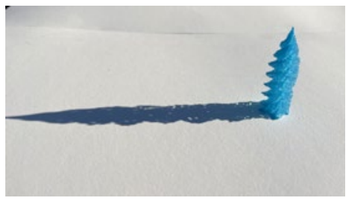


The many iterations:

Testing parameters: Angle of divergence, scale, feed rate, and printer.



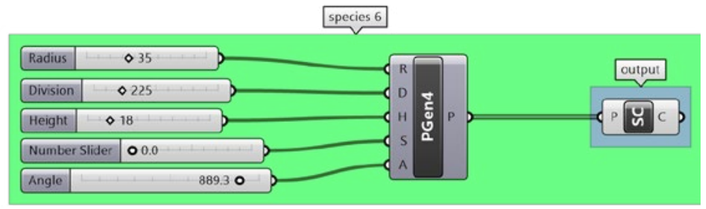


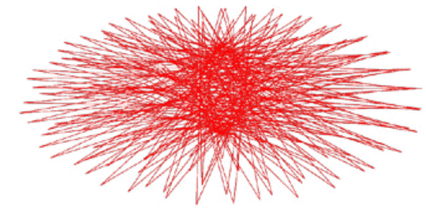


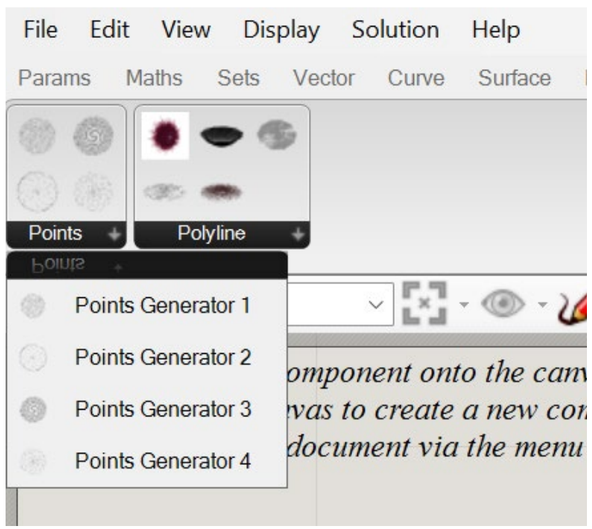
Model Name	Parameter: angle	Parameter: scale	Parameter: feed rate	Printer	Picture
final shell attempt	90	2	250	dremel	
3nd shell attempt	5	1	200	dremel	
2nd shell attempt	137.3	1	300	prusa	
1st shell attempt	137.3	1	300	dremel	
1st pine cone attempt	135.7	1	200	dremel	
2nd pine cone attempt	135.7	1	120	dremel	
final pine cone attempt	135.7	1	120	prusa	
1st icicle attempt	135.7	0.5	300	dremel	
final icicle attempt	135.7	0.5	250	prusa	
mini pine cones attempts	137.6	0.5	500	prusa	
sunflower center	137.5	1	300	prusa	
1st flower attempt	135.7	1	300	dremel	
transparent flower	135.7	1	120	dremel	
stem	n/a	0.5	300	dremel	

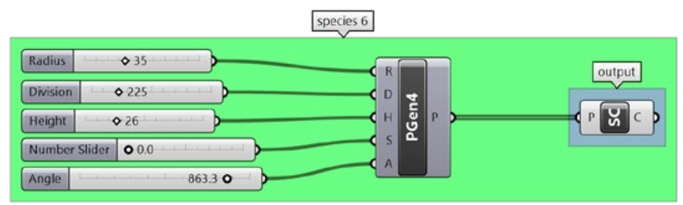
Simple & Beautiful:

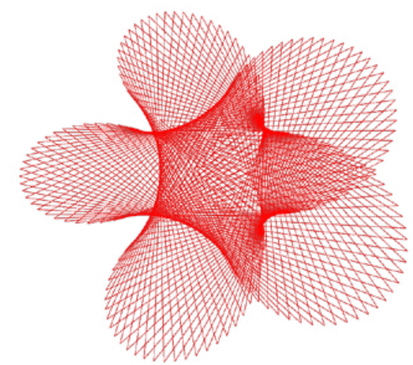
The plug-in includes four types of point generations, each following different angles of divergence, as well as four different species. The richness of shapes that one can achieve with these two configurations is endless.

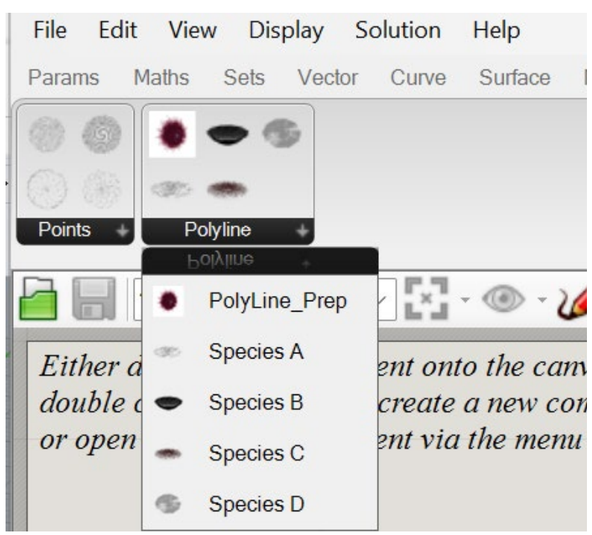


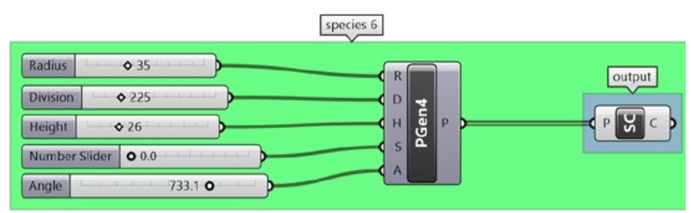


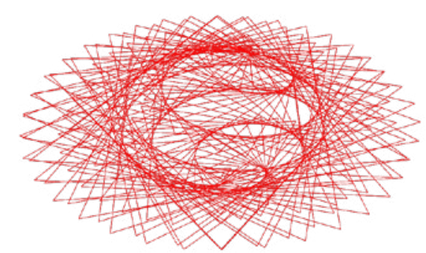










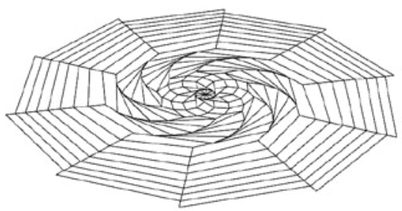


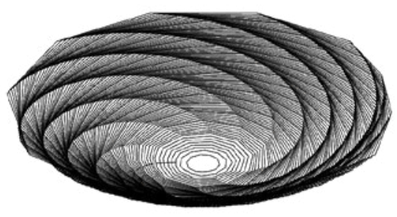
**Possibilities within a single species*

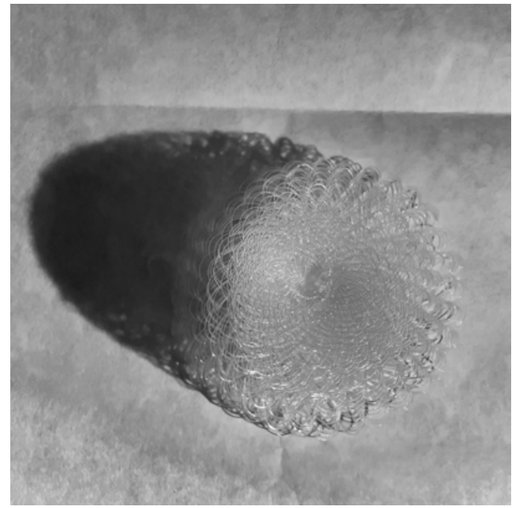
Food4Rhino + Documentation

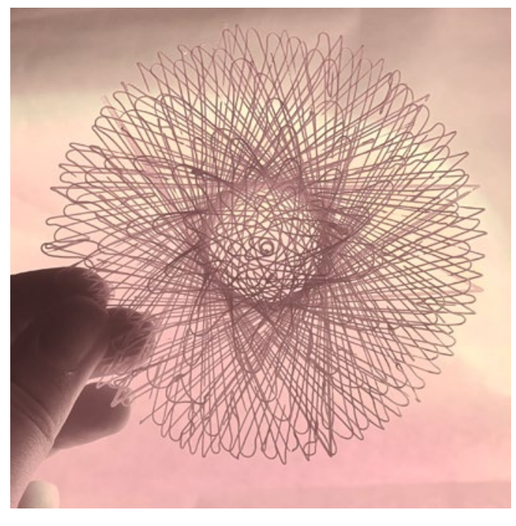
Visit :
https://gsd6338.org/fall2023/projects/seaurchin/index.html
for Harvard GSD documentation

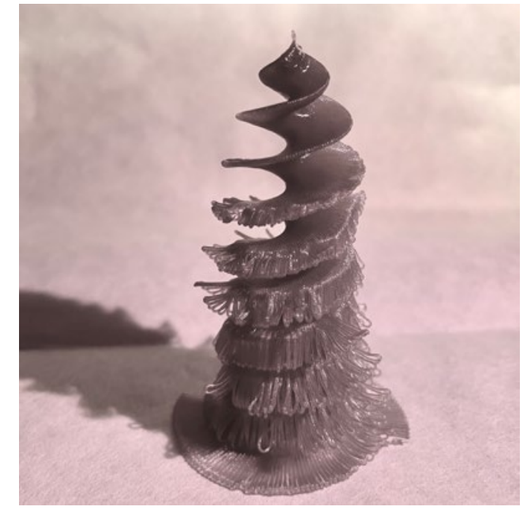
And:
https://www.food4rhino.com/en/app/seaurchin
for Food4Rhino download

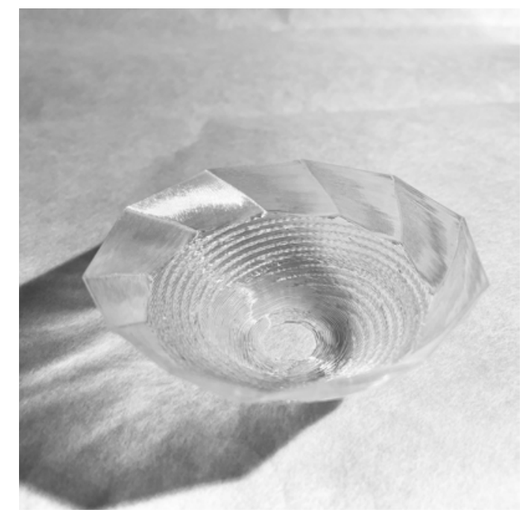


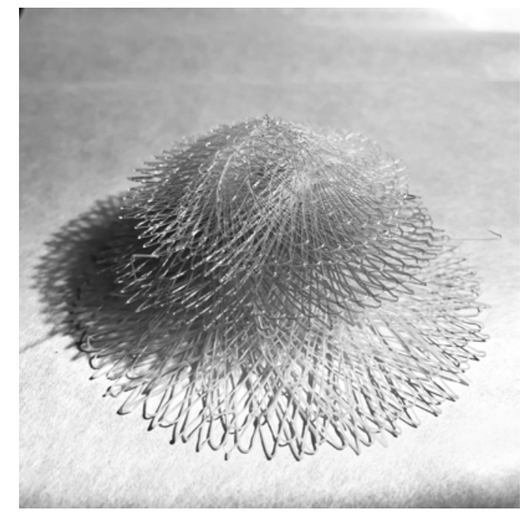












Thank you.

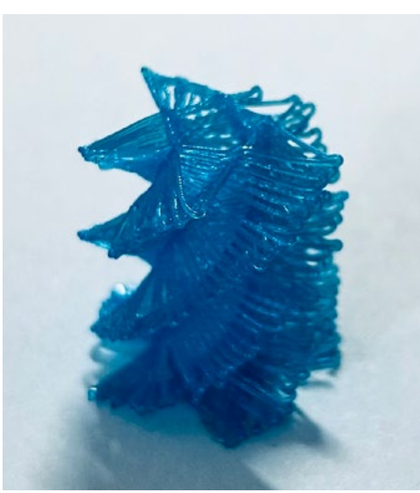
Design Challenge:
Stool Design

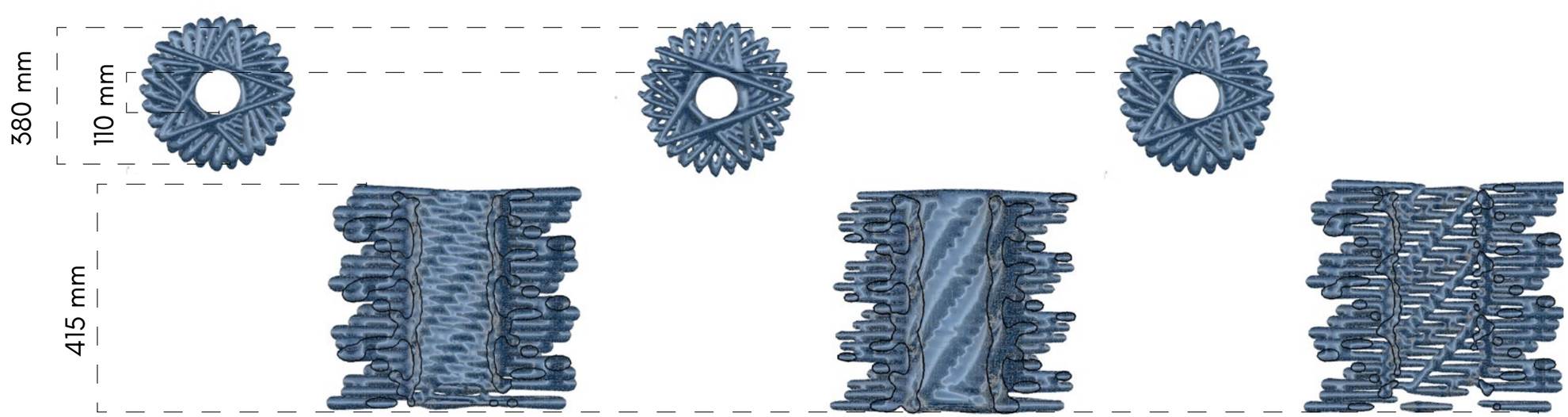
Exploring tree-like structure geomtries for:

Hierarchical strength: Natural optimization of material distribution.
Load transfer paths: Efficiently channels forces from many points to fewer.

Parameters:	Version 1: Touching layers.	Version 2: Edges Floating with stronger edges.	Version 3: Floating travel paths with connecting corners.
Bead Dimension	Varying bead dimension 6-8 mm for closer center layers 10-15 mm for edge	Single bead dimension of 8-10 mm	Varying bead dimension 10-15 mm for corners 6-8 mm for travel paths
Cross-Section shape	Oval and consistent	Oval and consistent	Variable geometry die to change aspect ration on travel paths (thiner on edges)
Air Pressure	Programmed changes: Lower in center Higher in edges	Programmed changes: Standard in center Lower in edges (for increased wall thickness stability)	Consistent air pressure (or if dial is kept consistant change air pressure to be higher on connecting points and lower on travel paths)

Overall Cross-Section:





Overall approach:	Expansion on edges for even support of all layers.	Decrease pressure on edges for stronger smaller edges (strong float).	Thin light and delicate travel paths with stronger support on edges.